Week 14 Assignment — Bootstrap Applied to a New Building

CE 310 | Individual submission | Due: Wednesday of Week 15 by 9:00 AM — submit to D2L

The lab bootstrapped the DOE Medium Office dataset you have used since Week 5. This assignment asks you to run the same methods on a building you have not seen, so that the numbers are genuinely new to you — a rehearsal for the capstone, where you will apply these tools to data nobody in the class has analysed before.

How to Submit

> ☐ Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
Run all cells	Runtime → Run all — wait until every cell shows a checkmark, no spinners
File format	Download as .ipynb — File → Download → Download .ipynb (not PDF, not .py)
Print lines	Do not edit or delete the print(f"ANSWER_A1 = ...") lines
Seed	Do not change SEED = 42 — results must use this exact seed
File name	lastname_firstname_week14.ipynb e.g. smith_jane_week14.ipynb
Submission	Upload the .ipynb file to D2L Dropbox before the deadline

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Dataset

Pick any one of the five building datasets below and use it for every problem. State which one you chose in the first cell of your notebook — your numbers will differ from your classmates', so your submission has to say which building they come from.

Week15_Hospital_Dataset.csv
Week15_PrimarySchool_Dataset.csv
Week15_LargeRetail_Dataset.csv
Week15_Warehouse_Dataset.csv
Week15_SmallHotel_Dataset.csv

All five share the DOE dataset's column structure — Outdoor_Temp_F, Cooling_kWh, BusinessHours (Yes/No) — so your lab code runs unchanged once you swap the filename.

The five buildings behave very differently: a hospital runs around the clock, a primary school empties at weekends and all summer, a warehouse has a large insulated thermal mass. Pick one whose operation you can reason about, because P3 and P4 ask you to explain the sign and size of its occupancy coefficient, not just report it.

> These are the same five datasets offered as the ready-made option for the Week 15 capstone. If your team plans to use one, this assignment is a useful head start — but you are not committed to it either way.

Tasks

P1. Bootstrap CI for the mean temperature (2 pts)
Using N_BOOT = 1000, seed = 42, compute the 95% percentile bootstrap CI for mean Outdoor_Temp_F. Compare to the analytical t-interval. Are they similar? (All buildings use the same Tucson outdoor temperatures, so expect the same CI as in the lab.)

P2. Bootstrap CI for the regression slope (4 pts)
Fit the simple model Cooling_kWh ~ Outdoor_Temp_F on your building's data. Compute the 1000-resample bootstrap 95% CI for the slope. Report [lower, upper]. How does it compare to the OLS analytical CI?

P3. Bootstrap CI for the BH coefficient (4 pts)
Fit Cooling_kWh ~ Outdoor_Temp_F + BH on your building's data. Compute the bootstrap 95% CI for the BH coefficient. Critical step: Does the CI include zero? If yes, what does that mean? If no, can you explain the sign and magnitude in terms of your building's operation?

P4. Written reflection (5 pts)
In 2–3 sentences: how does the bootstrap CI for BH compare to the OLS analytical CI for the same coefficient? Is the bootstrap CI wider, narrower, or the same? What might cause a difference?

P5. BONUS — Bootstrap CI for P(Cooling > P75) (bonus 3 pts)
Compute the 75th percentile of Cooling_kWh in your building's dataset. Then bootstrap the 95% CI for P(Cooling > P75). Report [lower, upper].

P6 — Permutation Test for BH Significance (3 pts)

Shuffle the BH column 1000 times. For each shuffle, refit Cooling_kWh ~ Outdoor_Temp_F + BH and record the BH coefficient. Report: - Your observed BH coefficient (from P3) -

The two-sided permutation p-value: fraction of shuffled coefficients whose absolute value exceeds |observed BH coefficient|

Does the permutation test agree with the OLS t-test conclusion from P3?

P7 — Bootstrap CI for R^2 (3 pts)

Using pairs bootstrap (same approach as P2), compute the 1000-resample percentile 95% CI for R^2 of the Cooling_kWh ~ Outdoor_Temp_F + BH model. Report [lower, upper].

Does the CI include $R^2 = 0.697$ (the DOE Medium Office value from Week 10)?

P8 — Written: Bootstrap Limitations (3 pts)

In 2–3 sentences, identify two situations in which the bootstrap confidence interval would give unreliable results for building energy data of this type. For each, explain why bootstrap fails there. (Examples to consider: autocorrelated data, biased or non-representative sample, very small n, extrapolating beyond the observed temperature range.)

Submission

Submit your Colab notebook (.ipynb) with all cells executed. Answers should appear in the cell outputs as ANSWER_P1_lower, ANSWER_P1_upper, ANSWER_P2_lower, ANSWER_P2_upper, ANSWER_P3_lower, ANSWER_P3_upper, and a written paragraph for P4.

Note: Because each student uses a different building, there is no single expected numeric answer for this assignment — your outputs will be reviewed individually.

Connection to Week 15

Your capstone team will run this same bootstrap step on whatever dataset it chooses, and the result belongs on Slide 2 (Regression Findings): report not just the coefficient but the uncertainty range around it. Getting the mechanics working here, on data whose behaviour you can already reason about, means the only new thing in Week 15 is the dataset.